

PRACTICAL_03

1. Develop a C program using `fork()` that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as `ps`, `top`, and `/proc`. Document the observations.

```
aasritha@DESKTOP-4B676UM:~/OS_LAB/Practical/Practical3$ ./Practical3

----- Parent Process -----

----- Child Process -----
Parent PID : 214
Child PID  : 215
Parent waiting for child...
Child PID  : 215
Parent PID : 214
Child is sleeping for 30 seconds...
Child process completed.
Child terminated.
Parent process completed.
aasritha@DESKTOP-4B676UM:~/OS_LAB/Practical/Practical3$ ps -ef | grep Practical3
aasritha  217   180  0 04:20 pts/0    00:00:00 grep --color=auto Practical3
aasritha@DESKTOP-4B676UM:~/OS_LAB/Practical/Practical3$ top
top - 04:20:24 up 34 min,  0 user,  load average: 0.00, 0.00, 0.00
Tasks:  5 total,   1 running,   4 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.1 us,  0.1 sy,   0.0 ni, 99.8 id,   0.0 wa,   0.0 hi,   0.0 si,   0.0 st
MiB Mem : 3889.3 total, 3706.6 free,   236.6 used,   118.8 buff/cache
MiB Swap: 1024.0 total, 1024.0 free,    0.0 used. 3652.8 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
218	aasritha	20	0	9276	5232	3088	R	0.7	0.1	0:00.02	top
1	root	20	0	1804	1184	1104	S	0.0	0.0	0:00.10	init
178	root	20	0	1812	84	0	S	0.0	0.0	0:00.00	init
179	root	20	0	1812	92	0	S	0.0	0.0	0:00.28	init
180	aasritha	20	0	6068	5368	3648	S	0.0	0.1	0:00.25	bash

```
aasritha@DESKTOP-4B676UM:~/OS_LAB/Practical/Practical3$ cat /proc/180/status
Name:   bash
Umask:  0022
State:  S (sleeping)
Tgid:   180
Ngid:   0
Pid:    180
PPid:   179
TracerPid: 0
Uid:    1000    1000    1000    1000
Gid:    1000    1000    1000    1000
FDSize: 256
Groups: 4 20 24 25 27 29 30 44 46 100 107 1000
NStgid: 180
NSpid:  180
NSpgid: 180
NSSid:  180
VmPeak: 6068 kB
VmSize: 6068 kB
VmLck:  0 kB
VmPin:  0 kB
VmHWM:  5368 kB
VmRSS:  5368 kB
RssAnon:          1720 kB
RssFile:          3648 kB
RssShmem:         0 kB
VmData: 1624 kB
VmStk:  132 kB
VmExe:  956 kB
VmLib: 1820 kB
VmPTE:  48 kB
VmSwap: 0 kB
HugetlbPages:    0 kB
CoreDumping: 0
```

```
RssAnon:          1720 kB
RssFile:          3648 kB
RssShmem:         0 kB
VmData:          1624 kB
VmStk:           132 kB
VmExe:           956 kB
VmLib:           1820 kB
VmPTE:            48 kB
VmSwap:           0 kB
HugetlbPages:     0 kB
CoreDumping:      0
THP_enabled:      1
Threads:          1
SigQ:            0/15540
SigPnd: 0000000000000000
ShdPnd: 0000000000000000
SigBlk: 0000000000010000
SigIgn: 0000000200384004
SigCgt: 000000004b813efb
CapInh: 0000000000000000
CapPrm: 0000000000000000
CapEff: 0000000000000000
CapBnd: 000001fffffffffff
CapAmb: 0000000000000000
NoNewPrivs:       0
Seccomp:          0
Seccomp_filters:  0
Speculation_Store_Bypass: thread vulnerable
Cpus_allowed:     f
Cpus_allowed_list: 0-3
Mems_allowed:     1
Mems_allowed_list: 0
voluntary_ctxt_switches: 476
nonvoluntary_ctxt_switches: 7
aasritha@DESKTOP-4B676UM:~/OS_LAB/Practical/Practical3$ |
```